

Customer Behaviour, Financial Risk, and Credit Default Analytics in a Retail Lending Environment

Background

A mid-sized Indian retail lending institution offers personal loans to salaried and self-employed customers across multiple cities. Over the past five years, the bank has accumulated large volumes of customer, transaction, and repayment data. While loan disbursement has grown steadily, management has raised concerns about rising defaults, fluctuating monthly performance, and unclear relationships between customer attributes and credit risk. The analytics team has been asked to examine historical data to understand customer behaviour, identify relationships among financial variables, analyse time-based performance patterns, and build a basic predictive model to assess credit default risk. As a junior business analyst, you are required to use Python to support managerial decision-making using data-driven insights.

Dataset Design

The capstone uses a single integrated dataset with approximately 8,000–10,000 rows, ensuring statistical stability and realism.

Key variable groups include:

- Customer demographics (age, gender, employment type, city tier)
- Financial attributes (monthly income, loan amount, EMI, tenure, interest rate)
- Behavioural indicators (credit utilisation, delayed payments, number of past loans)
- Time variables (loan start date, monthly observation period)
- Performance indicators (monthly repayment status, outstanding balance)
- Target variable (credit default: yes/no)

Task 1: Descriptive Statistics – Customer & Loan Profiling

Students begin by summarising customer and loan characteristics using descriptive statistics. They analyse central tendency, dispersion, and distribution shape to profile borrowers and loan behaviour. Emphasis is placed on **interpretation**, such as identifying variability in income, inconsistency in repayment behaviour, and concentration of loan exposure across customer segments. The objective is to convert numerical summaries into managerial understanding of the loan portfolio.

I. Dataset Overview Using `df.info()`

Brief Explanation of Commands Used in Jupyter

The `df.info()` command is used to generate a concise summary of the dataset, including the total number of observations, variable names, non-null counts, data types, and memory usage. This command helps in understanding the overall structure of the dataset and verifying whether the data is suitable for statistical analysis and modelling before proceeding to advanced analytical tasks.

Interpretation of Outputs and Their Business Implication

The output shows that the dataset contains **8,000 observations and 14 variables**, indicating a sufficiently large dataset for conducting descriptive, relational, and predictive analytics. All variables have **8,000 non-null values**, confirming that the dataset is complete and free from missing values. This ensures that subsequent business analyses, such as credit risk assessment and customer profiling, can be performed without the need for data cleaning or imputation at the initial stage.

The dataset consists of multiple data types, each serving a specific analytical purpose. **Integer (int64) variables** represent discrete numeric information such as age, income, loan amount, tenure, and repayment behaviour, which are essential for quantitative analysis and modelling. **Floating-point (float64) variables** capture continuous financial measures such as interest rate, credit utilisation, and EMI, allowing precise calculations and regression-based analysis. **Categorical (object) variables** represent qualitative customer attributes such as gender and employment type, enabling segmentation and comparative analysis across customer groups. Additionally, the presence of a **datetime (datetime64) variable** supports time-based analysis, making it possible to study trends and behavioural changes over different periods.

From a business perspective, this combination of data types enables comprehensive analysis of customer demographics, financial capacity, repayment behaviour, and temporal patterns. The dataset's efficient memory usage further indicates that it can be processed smoothly in a Jupyter Notebook environment, facilitating real-time analysis and classroom demonstrations. Overall, the dataset is well-structured and analytically robust for informed business decision-making.

II. Descriptive Statistics Using `df.describe()`

Brief Explanation of Commands Used in Jupyter

The `df.describe()` command is used to generate summary descriptive statistics for numerical variables in the dataset. It provides key measures such as count, mean, standard deviation, minimum value, quartiles, and maximum value. This command helps in understanding the

central tendency, dispersion, and distribution range of business-related numerical variables before conducting relational or predictive analysis.

Interpretation of Outputs and Their Business Implication

The descriptive statistics indicate that the dataset comprises **8,000 valid observations** across all selected numeric variables, confirming data consistency and completeness. The **average age of customers is approximately 40 years**, with values ranging from 21 to 59 years, suggesting that the loan portfolio primarily caters to working-age individuals. From a business perspective, this reflects a mature customer base with stable earning potential.

The **average monthly income is around ₹69,840**, with a substantial standard deviation, indicating wide income dispersion among customers. This variability suggests the presence of diverse income segments within the customer base, which has important implications for loan product differentiation and risk-based pricing strategies. Similarly, the **mean loan amount of approximately ₹7.71 lakh**, coupled with a broad range extending up to nearly ₹15 lakh, indicates a diversified loan portfolio catering to both low- and high-value borrowers.

The **average loan tenure of about 36 months** reflects a preference for medium-term loans, while the interest rate averages around **13.5%**, indicating standard pricing for retail lending products. The wide spread in EMI values, with a mean of approximately **₹38,398**, highlights significant variation in repayment obligations, which may influence affordability and default risk across different customer segments.

The **average credit utilisation ratio of 0.50** suggests moderate usage of available credit limits among customers. Higher utilisation levels observed at the upper quartiles indicate increased financial stress for certain borrowers, which may elevate credit risk. Additionally, the **average number of delayed payments is 2.5**, with some customers exhibiting up to five delays, signalling potential repayment behaviour concerns. From a business standpoint, these behavioural indicators are critical for early warning systems and credit risk assessment.

Overall, the descriptive statistics provide valuable insights into customer demographics, financial capacity, and repayment behaviour, forming a strong foundation for further analysis such as correlation, regression, and credit default prediction.

III. Variability and Distribution Shape Analysis

Brief Explanation of Commands Used in Jupyter

The aggregation function `agg(['var', 'skew', 'kurt'])` is used to compute additional descriptive statistics that go beyond basic summaries. Variance measures the spread or dispersion of data, skewness indicates the symmetry or asymmetry of a distribution, and kurtosis reflects the

peakedness or tail behaviour of the distribution. These measures help in understanding data distribution characteristics that are important for selecting appropriate analytical and modelling techniques.

Interpretation of Outputs and Their Business Implication

The variance values indicate substantial dispersion in key financial variables such as **monthly income, loan amount, and EMI**, highlighting considerable heterogeneity in customers' financial profiles. High variance in loan amount and EMI suggests that the institution caters to a wide spectrum of borrowers, ranging from low-value to high-value loans. From a business perspective, this diversity necessitates differentiated risk assessment and customised loan offerings rather than a uniform credit policy.

The skewness values for most variables are close to zero, indicating that the distributions of **age, income, loan amount, tenure, interest rate, credit utilisation, and delayed payments** are largely symmetric. This suggests a balanced distribution of customers across low and high values for these variables, which is favourable for statistical analysis and regression modelling. However, the **EMI variable exhibits a positive skew**, indicating the presence of a smaller group of customers with substantially higher repayment obligations. This asymmetry may signal higher affordability risk for certain borrowers and warrants closer monitoring.

The kurtosis values for most variables are negative, indicating relatively **flatter distributions** with lighter tails compared to a normal distribution. This implies fewer extreme outliers in variables such as income, loan amount, and credit utilisation, which enhances the stability of analytical results. In contrast, **EMI shows positive kurtosis**, suggesting a more peaked distribution with heavier tails. From a business standpoint, this points to the existence of extreme repayment values that could significantly influence default risk and should be factored into credit risk modelling and policy decisions.

Overall, the distributional characteristics suggest that while most variables are well-behaved and suitable for parametric analysis, the EMI variable deserves particular attention due to its higher variability and distributional asymmetry, reinforcing its importance in credit risk evaluation.

IV. Categorical Variable Distribution Analysis

Brief Explanation of Commands Used in Jupyter

The `value_counts(normalize=True)` function is used to calculate the relative frequency of categorical variables in percentage terms. By normalising the counts, this command helps in understanding the proportional distribution of different categories within the dataset. This is particularly useful for analysing customer composition and outcome balance in business datasets.

Interpretation of Outputs and Their Business Implication

The gender distribution in the dataset is nearly equal, with **male customers accounting for approximately 50.06%** and **female customers for 49.94%** of the total sample. This indicates a well-balanced gender representation in the loan portfolio, reducing the likelihood of gender-based bias in analysis and supporting fair and inclusive credit assessment practices.

In terms of employment type, **salaried customers constitute around 70.69%**, while **self-employed customers account for approximately 29.31%** of the dataset. This suggests that the institution's loan portfolio is predominantly driven by salaried individuals, who are typically perceived as having more stable income streams. From a business standpoint, this composition reflects a relatively conservative lending strategy, with controlled exposure to higher-risk self-employed segments.

The distribution of the credit default variable shows that **78.35% of customers are non-defaulters**, while **21.65% have defaulted**. This indicates a moderately imbalanced outcome variable, where non-default cases dominate the dataset. From a business perspective, this reflects a generally healthy loan portfolio, while still maintaining a sufficiently large proportion of default cases to support meaningful credit risk analysis and predictive modelling. The presence of both classes in reasonable proportions enables the development of effective classification models for default prediction.

Overall, the categorical distributions provide valuable insights into customer composition and portfolio health, which are critical inputs for segmentation, risk management, and policy formulation.

V. Employment Type and Financial Behaviour Analysis

Brief Explanation of Commands Used in Jupyter

The `groupby("Employment_Type").mean()` command is used to compute average values of selected numerical variables separately for each employment category. This command helps in comparing financial behaviour and loan characteristics across different customer segments, enabling segment-wise business insights.

Interpretation of Outputs and Their Business Implication

The group-wise analysis shows that the **average age of salaried and self-employed customers is identical at approximately 40 years**, indicating that both employment segments are demographically comparable. This suggests that age alone does not differentiate these two customer groups within the loan portfolio.

In terms of income, **self-employed customers exhibit a slightly higher average monthly income (₹70,667) compared to salaried customers (₹69,497)**. However, this difference is marginal and does not translate into significant variation in loan amount or EMI. Both groups have a similar **average loan amount of approximately ₹7.7 lakh** and comparable **EMI values of around ₹38,400**, indicating uniform credit exposure across employment types.

The **average loan tenure and interest rate** are also nearly identical for both segments, suggesting that the institution follows consistent pricing and tenure policies irrespective of employment category. Credit utilisation levels remain similar across groups, indicating comparable borrowing behaviour and financial discipline.

From a risk perspective, **self-employed customers show a marginally lower average number of delayed payments** compared to salaried customers. While the difference is small, it challenges the conventional assumption that self-employed borrowers are inherently riskier. From a business standpoint, this insight suggests the potential for more nuanced risk assessment models that go beyond employment type alone and incorporate behavioural indicators.

Overall, the analysis indicates that **employment type alone does not significantly influence financial behaviour or repayment characteristics** in this dataset, highlighting the importance of multi-factor credit evaluation rather than reliance on single demographic attributes.

VI. Credit Default Status and Financial Behaviour Analysis

Brief Explanation of Commands Used in Jupyter

The `groupby("Credit_Default").mean()` command is used to compute average values of selected numerical variables separately for defaulters and non-defaulters. This allows for a comparative assessment of financial capacity, borrowing behaviour, and repayment characteristics across default outcomes, supporting risk analysis and credit decision-making.

Interpretation of Outputs and Their Business Implication

The analysis reveals that the **average age of defaulters (40.55 years) and non-defaulters (40.07 years)** is nearly identical, indicating that age does not play a significant role in differentiating default behaviour in this dataset. This suggests that default risk is not strongly age-dependent and should not be assessed based on demographic factors alone.

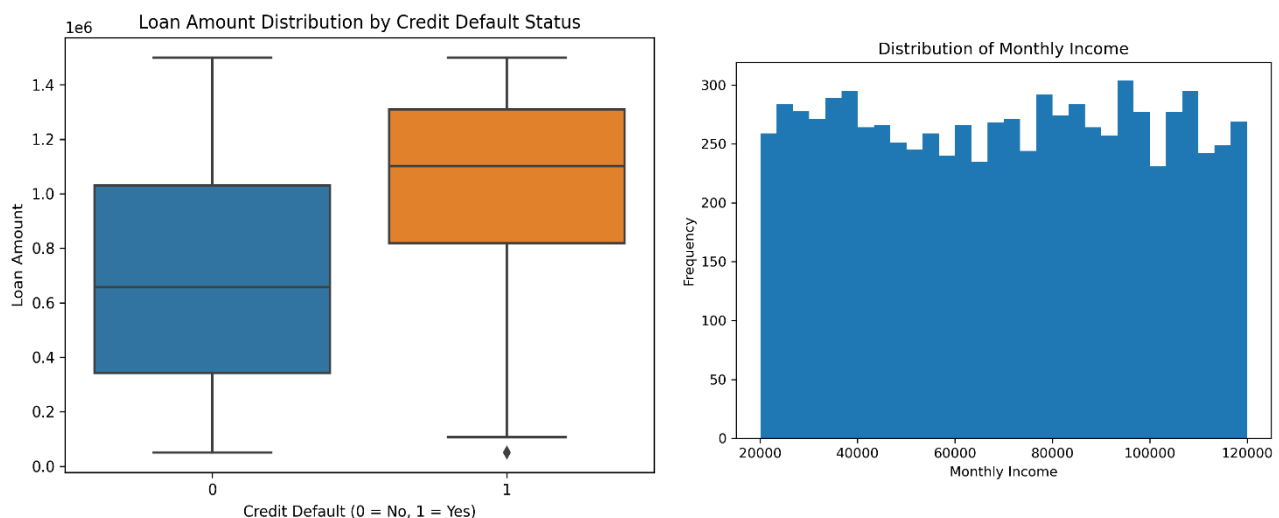
A notable difference is observed in **monthly income**, where **non-defaulters earn an average of ₹74,893**, while **defaulters earn significantly less at approximately ₹51,552**. This substantial income gap highlights income as a critical determinant of repayment capacity

and default risk. From a business perspective, lower income levels may constrain borrowers' ability to meet repayment obligations consistently.

The **average loan amount for defaulters is approximately ₹10.4 lakh**, which is considerably higher than the **₹6.97 lakh observed for non-defaulters**. Combined with a **shorter average loan tenure** for defaulters, this results in a significantly higher **average EMI of around ₹67,871**, compared to **₹30,254 for non-defaulters**. This indicates that defaulters face heavier repayment burdens relative to their income, increasing financial stress and default likelihood.

Additionally, defaulters exhibit **higher credit utilisation (0.61)** and **more delayed payments (3.30)** compared to non-defaulters, who show lower utilisation and fewer delays. These behavioural indicators signal weaker financial discipline and higher risk exposure among defaulters. From a business standpoint, such variables are valuable early warning indicators that can be incorporated into credit scoring and monitoring systems.

Overall, the results clearly demonstrate that **financial capacity and repayment behaviour**, rather than demographic characteristics, are the primary drivers of credit default. These insights provide strong justification for incorporating income, loan burden, credit utilisation, and payment history into predictive credit risk models.



VII. Visual Descriptive Analysis

Brief Explanation of Commands Used in Jupyter

A **histogram** was used to display the frequency distribution of the `Monthly_Income` variable across defined income intervals, helping to visualise how income values are spread within the dataset. A **boxplot** was used to compare the distribution of `Loan_Amount` between two groups defined by credit default status. The boxplot highlights the median, interquartile range, and

extreme values, enabling a visual comparison of loan exposure across defaulters and non-defaulters.

Interpretation of Outputs and Their Business Implication

The histogram of **monthly income** shows that income values are **widely and fairly evenly distributed across the full range**, from lower to higher income brackets. There is no sharp concentration at a single income level, indicating that the loan portfolio serves customers across diverse income segments. From a business perspective, this suggests that lending decisions are not restricted to a narrow income group, which increases market reach but also requires careful affordability assessment across varying income levels.

The boxplot comparing **loan amount by credit default status** shows a **clear visual separation between defaulters and non-defaulters**. The median loan amount for defaulters is visibly higher than that of non-defaulters, and the interquartile range for defaulters is wider. This indicates that customers who default tend to have **larger and more variable loan exposures**. Additionally, defaulters exhibit higher upper ranges and extreme values, suggesting greater financial burden.

From a business standpoint, these visual patterns reinforce the role of **loan size as a key risk driver**. Higher loan amounts appear to be associated with increased default risk, especially when not matched with proportionately higher income. This highlights the importance of incorporating loan amount thresholds, affordability ratios, and enhanced monitoring for high-value loans as part of credit risk management strategies.

Percentile Analysis of Key Financial Variables

Brief Explanation of Commands Used in Jupyter

The `quantile()` function is used to compute percentile values for numerical variables in the dataset. Percentiles divide the data into equal portions and help in understanding how values are distributed across different segments of the population. This command is particularly useful for identifying thresholds, segmenting customers, and analysing concentration of risk across lower, middle, and higher value groups.

Interpretation of Outputs and Their Business Implication

The percentile analysis shows a clear stratification of customers across demographic, financial, and behavioural dimensions. The **25th percentile age is 31 years**, the median age is **40 years**, and the **90th percentile reaches 56 years**, indicating that the loan portfolio largely serves mid-career individuals, with a smaller concentration of older borrowers at higher percentiles.

For **monthly income**, the 25th percentile is approximately **₹44,055**, while the median income is **₹70,364**, indicating a substantial income gap between lower and middle segments. At the upper end, the **90th and 95th percentiles exceed ₹1.09 lakh and ₹1.14 lakh respectively**, suggesting that a smaller proportion of customers earn significantly higher incomes. From a business perspective, this highlights the importance of income-based segmentation for product design and risk-based pricing.

The **loan amount distribution** shows that 50% of customers have loans up to **₹7.72 lakh**, while the top 25% exceed **₹11.27 lakh**, and the top 5% cross **₹14.26 lakh**. This indicates a strong concentration of credit exposure among higher-value borrowers. Such concentration increases portfolio risk and necessitates stricter appraisal and monitoring mechanisms for high-ticket loans.

Loan tenure percentiles indicate that **longer tenures dominate higher percentiles**, with the 90th and 95th percentiles both reaching **60 months**, suggesting that extended repayment periods are common among higher loan segments. Interest rates also rise steadily across percentiles, implying risk-adjusted pricing as loan size and tenure increase.

The **EMI percentiles** show a sharp escalation at higher levels, with the top 10% of customers paying EMIs above **₹78,722**, and the top 5% exceeding **₹1.09 lakh**. This steep increase reflects higher repayment pressure on a small segment of borrowers, which may elevate default risk. Similarly, **credit utilisation and delayed payments increase consistently across percentiles**, with the highest percentiles showing utilisation above **0.82** and delayed payments reaching the maximum observed value.

From a business standpoint, this percentile analysis clearly identifies **high-risk segments concentrated in the upper percentiles of loan amount, EMI, credit utilisation, and delayed payments**. These segments warrant focused risk controls, tighter affordability checks, and proactive monitoring to maintain portfolio health.

Task 2: Correlation and Covariance – Understanding Relationships

Building on the summaries, students examine relationships among financial variables such as income, loan amount, EMI, tenure, and credit utilisation. Correlation and covariance are used to identify how variables move together and whether relationships are positive, negative, strong, or weak. Students are guided to distinguish **statistical association from business relevance**, discussing implications for affordability assessment and risk monitoring.

Correlation Analysis of Financial Variables

Brief Explanation of Commands Used in Jupyter

The `corr()` function was used to compute the **Pearson correlation coefficients** between selected numerical variables. Correlation measures the **strength and direction of linear relationships** between pairs of variables, with values ranging from -1 to $+1$. This analysis helps identify which financial variables move together and which operate independently, supporting relationship analysis prior to regression modelling.

Interpretation of Outputs and Their Business Implication

The correlation results show that **monthly income has near-zero correlation with most variables**, including loan amount, tenure, EMI, credit utilisation, and delayed payments. This indicates that income levels alone do not strongly influence loan size, repayment structure, or repayment behaviour in this dataset. From a business perspective, this suggests that lending decisions are not strictly income-driven and may incorporate other criteria such as credit history and behavioural indicators.

A **strong positive correlation (0.70)** is observed between **loan amount and EMI**, indicating that higher loan amounts are associated with higher monthly repayment obligations. This relationship is expected and confirms the structural dependence of EMI on loan size. From a business standpoint, this reinforces the importance of monitoring loan amount carefully, as it directly influences repayment burden.

A **moderately strong negative correlation (-0.58)** is observed between **loan tenure and EMI**, indicating that longer loan tenures are associated with lower monthly instalments. This reflects standard lending practices where repayment pressure is reduced by extending tenure. From a credit policy perspective, this highlights tenure as a key lever for managing affordability and controlling short-term repayment stress.

Interest rate shows **very weak correlations** with all other variables, including EMI. This suggests that variations in interest rates within the dataset do not significantly influence monthly repayment amounts or borrower behaviour. From a business standpoint, this may indicate relatively narrow interest rate bands or standardised pricing across customer segments.

Credit utilisation and delayed payments show **near-zero correlation with most financial variables**, suggesting that repayment behaviour is not directly driven by loan size, tenure, or income alone. This highlights the importance of treating behavioural indicators as **independent risk signals**, rather than assuming they are consequences of financial exposure.

Overall, the correlation analysis indicates that while **structural relationships exist between loan amount, tenure, and EMI**, most other variables operate independently. This underscores the need for **multivariate analysis**, such as regression and classification models, rather than relying on simple pairwise relationships for credit risk assessment.

Covariance Analysis of Financial Variables

Brief Explanation of Commands Used in Jupyter

The `cov()` function is used to compute the **covariance between pairs of numerical variables**. Covariance indicates the **direction of joint movement** between variables—whether they tend to increase or decrease together—but not the strength of the relationship in standardized terms. Unlike correlation, covariance values depend on the scale of the variables and are therefore interpreted mainly in terms of direction rather than magnitude.

Interpretation of Outputs and Their Business Implication

The covariance results show that **monthly income has a negative covariance with EMI**, indicating that higher income levels tend to be associated with lower repayment pressure. From a business standpoint, this suggests that customers with stronger income capacity are better positioned to manage monthly obligations, reinforcing income as an affordability indicator.

A **large positive covariance between loan amount and EMI** is observed, confirming that increases in loan size are accompanied by increases in monthly instalments. This reflects the structural relationship between borrowing amount and repayment burden. In practical terms, higher loan exposure directly translates into higher cash outflows, which can increase financial stress if not aligned with income levels.

Loan tenure shows a **negative covariance with EMI**, indicating that longer repayment periods are associated with lower monthly instalments. This supports the role of tenure as a key policy lever used by lenders to reduce short-term repayment pressure and improve affordability for borrowers.

Interest rate exhibits relatively small covariance values with other variables, including EMI. This suggests that variations in interest rates within the dataset do not materially drive repayment burden or behavioural outcomes. From a business perspective, this may indicate standardized pricing practices or narrow interest rate bands across customers.

Credit utilisation and delayed payments show **low covariance values with most financial variables**, indicating that repayment behaviour and credit usage patterns do not move mechanically with income, loan size, or tenure. This highlights their role as **behavioural risk**

indicators, which capture borrower discipline and financial stress beyond basic loan characteristics.

Overall, the covariance analysis reinforces that **loan amount and tenure directly influence repayment burden**, while behavioural variables such as credit utilisation and delayed payments operate more independently. These insights justify the use of multivariate models, such as regression and logistic classification, to capture combined effects in credit risk assessment rather than relying on pairwise relationships alone.

Simple Linear Regression: Loan Amount → EMI

Brief Explanation of Commands Used in Jupyter

A **simple linear regression model** was fitted using the `LinearRegression()` class from the sklearn library, where **Loan Amount** was treated as the independent variable and **EMI** as the dependent variable. The model estimates two key parameters: the **intercept (constant)** and the **slope (regression coefficient)**, which together define the linear relationship between the variables.

Interpretation of Outputs and Their Business Implication

The **intercept value is -809.65**, which represents the estimated EMI when the loan amount is zero. From a practical business perspective, this value does not have direct real-world meaning, as a loan amount of zero is not a valid lending scenario. However, it serves as a mathematical baseline required to define the regression line.

The **slope (regression coefficient) is 0.0508**, indicating that for every **₹1 increase in loan amount**, the EMI increases by approximately **₹0.05**, assuming other factors remain constant. In practical terms, this means that an increase of **₹1,00,000 in loan amount** would raise the EMI by approximately **₹5,083**, highlighting a clear and direct relationship between loan size and repayment burden.

From a business standpoint, this result confirms that **loan amount is a strong driver of EMI**, reinforcing the importance of aligning loan size with the borrower's repayment capacity. The positive coefficient indicates that higher loan exposure directly increases monthly financial obligations, which may elevate default risk if not carefully assessed during credit appraisal. This insight supports the use of EMI affordability checks and loan amount caps as essential components of credit risk management.

Model Fit Assessment Using R-Squared

Brief Explanation of Commands Used in Jupyter

The `model.score(X, y)` command is used to compute the **R-squared value** of the regression model. R-squared measures the proportion of variance in the dependent variable that is explained by the independent variable in a regression model. Its value ranges from 0 to 1, with higher values indicating better explanatory power.

Interpretation of Outputs and Their Business Implication

The R-squared value obtained is **0.495**, indicating that approximately **49.5% of the variation in EMI** is explained by **loan amount alone** in the simple linear regression model. This suggests a **moderate level of explanatory power**, confirming that loan amount is an important factor influencing EMI but not the sole determinant.

From a business perspective, this result implies that while loan size plays a significant role in determining repayment burden, **nearly half of the variation in EMI is influenced by other factors** such as loan tenure, interest rate, and borrower characteristics. Therefore, relying only on loan amount to assess repayment capacity would be insufficient.

This finding justifies the use of **multiple regression models** in credit analytics, where additional variables are incorporated to improve explanatory accuracy and support more informed lending decisions. It also reinforces the need for a holistic evaluation framework that combines loan characteristics with borrower-specific factors.

Simple Time Series Analysis of Credit Default Rate

Brief Explanation of Commands Used in Jupyter

The dataset was first aggregated using the `groupby()` function to calculate the **average credit default rate for each month**. The time variable was set as the index to enable time series analysis. A line plot was then used to visualise changes in default rates over time. Additionally, a **rolling mean with a window of three months** was computed using the `rolling().mean()` function to smooth short-term fluctuations and highlight underlying trends. The `describe()` function was applied to summarise the statistical properties of both the original series and the rolling mean.

Interpretation of Outputs and Their Business Implication

The monthly credit default rate values indicate that default behaviour **varies over time**, with observed monthly rates ranging from approximately **0.144 to 0.281**. The mean default rate across the entire period is around **0.217**, suggesting that, on average, about **21–22% of**

customers default each month. The standard deviation of approximately **0.032** indicates moderate variability in default behaviour over time rather than extreme volatility.

The rolling mean series shows a smoother pattern, with reduced variability compared to the original monthly values. The rolling mean ranges from approximately **0.183 to 0.263**, and its lower standard deviation confirms that short-term fluctuations are being averaged out. This smoothing helps reveal the underlying movement of default rates without being distracted by month-to-month noise.

From a business perspective, the presence of **visible fluctuations rather than a flat pattern** indicates that credit default behaviour is **time-dependent** and influenced by external or cyclical factors such as economic conditions, seasonal spending patterns, or changes in borrower behaviour. The relatively stable average level, combined with periodic increases and decreases, suggests that default risk is persistent but not erratic.

The rolling mean analysis is particularly useful for management, as it provides a clearer view of **medium-term default trends**, enabling better forecasting and proactive risk management. These results justify the need for time-aware credit monitoring systems rather than relying solely on static, cross-sectional analysis.